VTU-ETR Seat No. | | | | | | |

Question Paper Version : A

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) October - 2025

DO NOT USE MOBILE PHONES

Computer Science and Engineering/Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
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14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

Key Answers (CS/IS): **Version A**

PART - I

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	**b**	11	**b**	21	**b**	31	**a**	41	**b**
2	**b**	12	**c**	22	**c**	32	**b**	42	**a**
3	**b**	13	**c**	23	**a**	33	**b**	43	**a**
4	**c**	14	**d**	24	**c**	34	**a**	44	**b**
5	**d**	15	**b**	25	**b**	35	**b**	45	**b**
6	**c**	16	**b**	26	**c**	36	**b**	46	**c**
7	**c**	17	**b**	27	**b**	37	**c**	47	**b**
8	**c**	18	**a**	28	**b**	38	**b**	48	**c**
9	**b**	19	**b**	29	**b**	39	**d**	49	**c**
10	**b**	20	**b**	30	**b**	40	**c**	50	**c**

PART – II

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
51	b	61	**b**	71	a	81	b	91	a
52	b	62	**d**	72	d	82	a	92	b
53	c	63	**d**	73	d	83	**a**	93	d
54	c	64	**a**	74	b	84	c	94	c
55	c	65	**b**	75	d	85	a	95	b
56	c	66	**c**	76	c	86	d	96	b
57	b	67	**d**	77	b	87	b	97	b
58	a	68	**b**	78	c	88	b	98	a
59	d	69	**b**	79	c	89	b	99	a
60	a	70	**a**	80	b	90	c	100	b

PART – I

(Research Methodology)

[50 Marks]

1. What is the primary purpose of research?
a) To summarize existing knowledge
b) To discover new facts or verify old facts
c) To promote a specific political agenda
d) To create complex statistical models

2. Which of the following best describes 'Applied Research'?
a) Research carried out to generate knowledge for its own sake.
b) Research aimed at finding a solution to an immediate problem
c) Research that only uses qualitative methods
d) Research based purely on historical data.

3. The first and most critical step in the research process is:
a) Formulating a hypothesis
b) Defining the research problem
c) Collecting data
d) Preparing the research report

4. A hypothesis that indicates a direction of the expected difference or relationship is known as a:
a) Null hypothesis
b) Statistical hypothesis
c) Directional hypothesis
d) Complex hypothesis

5. Which research design is primarily used to establish cause-and-effect relationships?
a) Descriptive research design
b) Exploratory research design
c) Diagnostic research design
d) Experimental research design

6. What is the primary purpose of a literature review in research?
a) To show off how many books you have read
b) To avoid having to do original research
c) To gain insights into the existing theoretical and research background of the problem
d) To copy the methodology of previous studies exactly

7. A literature review should primarily be based on:
a) Personal opinions and anecdotes
b) Secondary sources like textbooks and review articles
c) Primary sources like original research reports and journal articles
d) Unverified online blogs and websites

8. Which of the following is a key characteristic of a good literature review?
a) It is highly subjective and opinionated.
b) It is a simple list of summaries of various articles.
c) It is comprehensive, critical, and synthesizes existing work.
d) It only includes studies that support the researcher's preconceived notions.

9. A 'secondary source' in a literature review refers to:
a) The second most important source you find
b) An interpretation or analysis of primary sources by other authors
c) The original reporting of research by the person who conducted it
d) A source that is less than 5 years old

10. The 'gap in literature' that a researcher identifies refers to:
a) A printing error in a journal article
b) An area that has not been researched or is underexplored
c) The time gap between two published studies
d) The margin space on a printed page

11. The main objective of a sample survey is to:
a) Collect data from every single member of the population
b) Collect data from a subset of the population to make inferences about the whole population
c) Only describe the characteristics of the sample itself
d) Replace the need for any statistical analysis

12. Which of the following is a probability sampling method?
a) Quota Sampling   b) Judgmental Sampling
c) Simple Random Sampling   d) Convenience Sampling

13. A sampling frame is:
a) The physical frame around a sample
b) The size of the sample
c) A list of all the elements in the population from which the sample is drawn
d) The margin of error in a survey

14. If a researcher stands in a shopping mall and selects shoppers who appear friendly to complete a survey, this is an example of:
a) Stratified Sampling   b) Systematic Sampling
c) Snowball Sampling   d) Convenience Sampling

15. The difference between the sample result and the true population value is known as:
a) Standard Deviation   b) Sampling Error
c) Non-sampling Error   d) Measurement Error

16. The process of checking completed questionnaires for accuracy, consistency, and completeness is called:
a) Data Analysis   b) Data Editing
c) Data Coding   d) Data Tabulation

17. Assigning numerical codes to response categories for data analysis is known as:
a) Data Editing   b) Data Coding
c) Data Entry   d) Data Validation

18. Which of the following is a method to handle missing data?
a) Ignore all cases with any missing values (listwise deletion)
b) Fabricate data to fill the gaps
c) Always replace missing values with the mean
d) There is only one correct method

19. The primary goal of data preparation is to:
a) Change the data to support the desired hypothesis
b) Ensure the data is accurate, consistent, and ready for analysis
c) Make the data look more impressive
d) Shorten the dataset by removing variables

20. Creating a frequency distribution table is part of which stage?
a) Data Collection b) Data Presentation
c) Hypothesis Testing d) Report Writing

21. A measure describing a characteristic of a population is called a:
a) Statistic b) Parameter
c) Sample d) Estimate

22. The Central Limit Theorem states that as sample size increases, the sampling distribution of the mean:
a) Becomes more skewed
b) Approaches a binomial distribution
c) Approaches a normal distribution, regardless of the population's distribution
d) Becomes identical to the population distribution

23. The standard deviation of the sampling distribution of a statistic is called its:
a) Standard Error b) Sampling Bias
c) Confidence Interval d) Population Deviation

24. An interval estimate, stated with a specific level of confidence, is called a:
a) Point Estimate b) Parameter Estimate
c) Confidence Interval d) Standard Interval

25. The purpose of statistical inference is to:
a) Describe the data in a sample
b) Make generalizations about a population from a sample
c) Collect data through surveys
d) Present data in graphical form

26. A hypothesis that is tested for possible rejection under the assumption that it is true is called the:
a) Alternative hypothesis b) Research hypothesis
c) Null hypothesis d) Directional hypothesis

27. The probability of rejecting a null hypothesis when it is actually true is known as:
a) Type II error (β) b) Level of Significance (α)
c) Power of the test ($1-\beta$) d) p-value

28. ANOVA is primarily used to test hypotheses about the differences among:
a) The variances of two populations
b) The means of two or more populations
c) The medians of two populations
d) The standard deviations of three populations

29. Which test is appropriate to compare means of two independent groups when population variances are equal and sample size moderately large?
a) Paired t-test
b) Independent samples t-test (Student's t)
c) Chi-square test
d) Wilcoxon signed-rank test

30. The "decision-making perspective" of business research emphasizes:
a) Maximizing the academic contribution of research
b) Reducing uncertainty for managers by providing relevant information
c) Collecting large amounts of data irrespective of its relevance
d) Promoting statistical software use in organizations

31. Which of the following illustrates a symptom rather than the true research problem?
a) Declining sales in a retail chain
b) Poor consumer perception of product quality
c) Ineffective promotional strategy
d) Limited market expansion opportunities

32. Which principle of the scientific method is most directly violated if a researcher hides unfavorable results?
a) Replication
b) Objectivity
c) Empiricism
d) Parsimony

33. .Which of the following is considered an ethical responsibility of a business researcher?
a) Guaranteeing positive results for the client
b) Maintaining respondent confidentiality
c) Concealing the research methodology to protect proprietary techniques
d) Ensuring sponsors dictate all research questions

34. Which of the following is an example of secondary data?
a) A company's sales reports from last year
b) A new survey of customer satisfaction
c) A focus group discussion with consumers
d) An experiment testing price sensitivity

35. If an ANOVA test yields a significant result, it indicates that:
a) All group means are significantly different from each other.
b) At least one group mean is significantly different from the others.
c) The variances of the groups are equal.
d) The null hypothesis of equal means is accepted.

36. In simple random sampling (SRS) without replacement, increasing sample size will generally:
a) Increase standard error of an estimate
b) Decrease standard error of an estimate
c) Not affect standard error
d) Make estimates invalid

37. Which of these is NOT a characteristic of good research?
a) Systematic
b) Replicable
c) Biased conclusions to support the investigator
d) Logical and empirical

38. The theorem that revises prior probabilities based on new evidence is:
a) Central Limit Theorem
b) Bayes' Theorem
c) Theorem of Total Probability
d) Pythagoras Theorem

39. Which of the following is a necessary condition for a Binomial distribution?
a) The number of trials is variable.
b) Each trial has more than two possible outcomes.
c) The probability of success changes from trial to trial.
d) Trials are independent.

40. A situation where you count the number of heads in 20 coin tosses is best modeled by a:
a) Normal distribution
b) Poisson distribution
c) Binomial distribution
d) Uniform distribution

41. The process of converting a normal random variable X to a standard normal variable Z is called:
a) Normalization
b) Standardization
c) Variance reduction
d) Centralization

42. In a Poisson distribution, the mean is equal to the:
a) Variance
b) Standard Deviation
c) Square of the variance
d) Probability of success

43. In mixed-method research design, which of the following best represents the "sequential explanatory strategy"?
a) Quantitative data collection followed by qualitative analysis for exploration
b) Qualitative exploration embedded within a quantitative experiment
c) Concurrent collection of qualitative and quantitative data with equal weight
d) Iterative cycles of qualitative and quantitative methods without fixed precedence

44. Exploratory research is usually conducted when:
a) The problem is well defined
b) The researcher has little knowledge about the problem
c) The study is about testing a clear hypothesis
d) The goal is to make accurate forecasts

45. In two-stage cluster sampling, the design effect (DEFF) is most directly influenced by:
a) Population mean
b) Intra-class correlation coefficient
c) Sampling frame completeness
d) Nonresponse rate

46. A statement such as, "If we increase our online advertising budget by 20%, then website traffic will increase by 15%" is an example of a:
a) Research objective
b) Management dilemma
c) Hypothesis
d) Research variable

47. The primary difference between a management dilemma and a research problem is that:
a) A management dilemma is faced by the researcher, while a research problem is faced by the manager.
b) A management dilemma is a symptom of an opportunity or problem, while a research problem is a statement about what information is needed to address it.
c) A management dilemma is always ethical in nature, while a research problem is methodological.
d) They are synonymous terms and can be used interchangeably.

48. A researcher is testing a new training program's effect on productivity. The possibility that the participants' awareness of being in an experiment itself influences their productivity is known as:
a) The halo effect
b) A contingency effect
c) An interactive testing effect
d) A surrogate information error

49. In the statement, "Brand loyalty is measured by the proportion of a customer's annual purchases in a category that are made from our brand," the phrase "proportion of purchases" serves as the:
a) Conceptual definition
b) Theoretical construct
c) Operational definition
d) Empirical scheme

50 A major fast-food chain wants to research the feasibility of launching a new plant-based burger nationwide. They decide to first conduct small-scale taste tests and focus groups in three cities. This initial study is an example of:
a) A conclusive research design
b) A cross-sectional study
c) Exploratory research
d) A performance-monitoring analysis

PART – II
Discipline Oriented Section
(Computer Science and Engineering/Information Science and Engineering)

[50 MARKS]

51. What is a major drawback of using recursion for problems that can also be solved iteratively?
a) Recursion always results in slower execution time
b) The risk of a stack overflow error for large inputs
c) The code is often more difficult to read and understand
d) It cannot handle problems with multiple base cases

52. In a linked list, what is the best approach to handle a memory leak caused by a failed deletion?
a) Not so important as modern systems handle memory management automatically
b) Use a 'free()' call on the node's pointer after it has been removed from the list's structure
c) Set the node's pointer to 'NULL' to indicate that it is no longer used
d) Introduce a garbage collection algorithm to track unreferenced nodes

53. A binary search tree is generated by inserting in the order of the following integers (from left to right). 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is
a) 4, 7 b) 8, 3 c) 7, 4 d) 6, 5

54. What will be the output of the following code?

```
int mystery(int n) {
   if (n <= 0) {
      return 0;
   } else {
      return n + mystery(n - 2);
   }
}

int main( ) {
printf("%d", mystery(5));
   return 0;
}
```

a) 0 b) 5 c) 9 d) 15

55. What is the maximum number of nodes in a binary tree of height 'h'?
a) 2h b) 2^h c) $2^{h+1}-1$ d) h^2

56. The truth values of the Propositions p, q and r if P →(Q V R) is False
a) P is True, Q is True and R is False
b) P is False, Q is False and R is True
c) P is True, Q is False and R is False
d) P is False, Q is True and R is True

57. How many integers between 1 and 300 (inclusive both) are divisible by at least one of 5, 6, 8?
a) 180 b) 120 c) 240 d) 138

58. In a directed Graph, there exists path between every pair of vertices, and then the Graph is called
a) Strongly connected Graph
b) Complete Graph
c) Isomorphic Graph
d) Euler Graph

59. A box contains 15 IC chips, of which 7 are defective and 8 are non- defective. In how many ways 5 chips can be chosen so that 2 are non-defective?
a) 56 b) 21 c) 1176 d) 980

60. Which one of the following Relation R: A-> A is not a function if A={1, 2, 3, 4}
a) R1={ (1, 2), (2, 4), (1, 3), (3, 4)}
b) R2={ (1, 1), (2, 2), (3, 3), (4, 4)}
c) R3={ (1, 3), (2, 4), (3, 4), (4, 1)}
d) R4={(1, 3), (2, 4, (3, 3), (4, 4)}

61. Which of the following best defines software engineering?
a) The process of writing code to solve problems
b) Engineering discipline concerned with all aspects of software production
c) Management of software projects
d) None of the above

62. Which of the following is not one of the four major phases of RUP?
a) Inception b) Elaboration c) Construction d) Integration

63. In software project estimation, which of the following is not a valid size estimation technique?
a) Lines of Code (LOC)
b) Function Point Analysis
c) Use-Case Points
d) Data Flow Diagrams

64. What is the difference between functional and non-functional requirements?
a) Functional requirements define what the system should do, while non-functional requirements define how the system should behave
b) Functional requirements define how the system should behave, while non-functional requirements define what the system should do
c) Functional requirements are related to performance, while non-functional requirements are related to security
d) Functional requirements are related to security, while non-functional requirements are related to performance

65. In UML, aggregation represents:
a) Strong ownership (like inheritance)
b) A weak "whole-part" relationship where parts can exist independently
c) A dynamic interaction
d) Implementation of interfaces

66. Which of the following addressing modes allows the program to access a memory operand whose address is obtained by adding a constant value to the contents of a register?
a) Immediate addressing
b) Direct addressing
c) Indexed addressing
d) Register indirect addressing

67. Which of the following statements about cache memory is TRUE?
a) Cache memory is slower than main memory but cheaper
b) Cache memory uses magnetic storage technology
c) Cache memory eliminates the need for virtual memory
d) Cache memory stores a copy of a portion of main memory

68. A hardwired control unit is generally:
a) More flexible but slower than micro-programmed control
b) Faster but less flexible than micro-programmed control
c) Both slower and less flexible than micro-programmed control
d) Faster and more flexible than micro-programmed control

69. Direct Memory Access (DMA) improves performance because:
a) It reduces memory access time for the CPU
b) It allows I/O devices to access memory directly without CPU intervention
c) It eliminates the need for interrupts
d) It bypasses both CPU and memory in data transfer

70. In floating-point representation (IEEE single precision), the bias value for the exponent is:
a) 127 b) 63 c) 255 d) 1023

71. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is
a) $O(n^2)$ b) O (n long n) c) $O(n^2 \log n)$ d) $O(n^3)$

72. Which of the following is/are property/properties of a dynamic programming problem?
a) Optimal substructure
b) Overlapping sub problems
c) Greedy approach
d)Both optimal substructure and overlapping sub problems

73. What is the purpose of asymptotic notation in algorithm analysis?
a) To precisely measure the exact number of operations in an algorithm
b) To provide a rough estimate of algorithmic performance
c) To compare algorithms based on their exact execution times
d) To simplify the comparison of algorithms based on their growth rates

74. Which of the following algorithms is the best approach for solving Huffman codes?
a) Exhaustive Search
b) Greedy Algorithm
c) Brute Force Algorithm
d) Divide and Conquer Algorithm

75. Which of the following problems is NOT solved using dynamic programming?
a) 0/1 knapsack problem b) Matrix chain multiplication problem
c) Edit distance problem d) Fractional knapsack problem

76. Which of the following is an advantage of multi-threaded programming?
a) Increases context-switch overhead
b) Decreases responsiveness of applications
c) Improves responsiveness and resource sharing
d) Eliminates synchronization needs

77. A multi-threaded program has 5 user-level threads mapped to 2 kernel-level threads in a many-to-many model. If two threads issue blocking I/O calls, how many threads can still execute?
a) 5 b) 3 c) 2 d) 0

78. A process has a working set of 10 pages. The physical memory allocated is 8 frames. If the program accesses all 10 pages frequently:
a) No page faults will occur
b) Page faults may occur due to demand paging
c) Thrashing is likely to occur
d) The process will be terminated

79. Which of the following is true about user-level threads vs kernel-level threads?
a) User-level threads are slower to create and manage than kernel-level threads
b) Kernel-level threads are managed by the user-level library
c) User-level threads are scheduled by the user-level thread library
d) Kernel-level threads are invisible to the OS

80. Which of the following is stored in a page table?
a) Virtual address b) Physical frame number
c) Page fault count d) Process ID

81. In digital transmission, what is the major limitation of increasing baud rate beyond the channel bandwidth?
a) Signal attenuation increases linearly
b) Inter-symbol interference increases due to limited bandwidth
c) Noise decreases, improving signal quality
d) The data rate no longer applies

82. Which mechanism at the Transport Layer ensures reliability over an unreliable Network Layer?
a) Flow control and sequencing b) ARP and ICMP messages
c) Error detection at the physical layer d) CSMA/CD protocol

83. Which of the following is a characteristic of wired LANs such as Ethernet?
a) Uses CSMA/CD for collision detection b) Uses token passing for media access
c) Provides wireless connectivity d) Requires satellites for communication

84 Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer
c) Presentation layer d) Transport layer

85. Which of the following is true about the role of ARP (Address Resolution Protocol) in network communication?

a) ARP translates IP addresses to MAC addresses to facilitate communication on a LAN

b) ARP translates domain names to IP addresses

c) ARP is used to detect errors in transmitted packets

d) ARP provides routing information between different networks

86. A retail store system has classes Customer, Order, and Payment. If one Order must be linked to exactly one Payment, but a Customer can place many Orders, which multiplicities are correct?

a) Customer 1..* → Order 1..* ; Order 1..* → Payment 0..1

b) Customer 1 → Order 1..* ; Order 1 → Payment 1

c) Customer 1..* → Order 0..* ; Order 1 → Payment 0..*

d) Customer 1 → Order 0..* ; Order 1 → Payment 1

87. Which of the following best distinguishes use case diagrams from class diagrams in Object-Oriented Modelling?

a) Use case diagrams emphasize system structure while class diagrams emphasize system behavior

b) Use case diagrams emphasize functional requirements, while class diagrams emphasize static structure

c) Use case diagrams and class diagrams both model interactions with actors

d) Use case diagrams cannot describe relationships such as generalization

88. In a **ride-hailing system**, a *"Book Ride"* use case always requires *"Select Pickup Location"* and *"Select Drop Location"*, but only sometimes involves *"Apply Promo Code"*. Which UML modeling best represents this?

a) Extend for Pickup and Drop, Include for Promo Code

b) Include for Pickup and Drop, Extend for Promo Code

c) Generalization of all three

d) Association only

89. During use case realization, which diagram is typically used to **show the flow of messages between objects**?

a) Class Diagram b) Sequence Diagram

c) State Diagram d) Deployment Diagram

90. Which design pattern category does the *Observer pattern* belong to, and why?

a) Creational, because it dynamically generates observers on demand

b) Structural, because it defines object composition

c) Behavioral, because it manages interaction and communication among objects

d) Architectural, because it defines system-wide layering

91. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____integrity constraint.

a) Referential b) Primary c) Referencing d) Specific

92. Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering
a) I only b) (B) II only c) Both I and II d) Neither I nor II

93. Which of the following concurrency control techniques does not ensure conflict-serializability?
a) Two-phase locking b) Timestamp ordering
c) Validation-based protocols d) Strict timestamp ordering with cascading aborts

94. Which of the following problems can still occur under the Read Committed isolation level?
a) Dirty read b) Lost update
c) Non-repeatable read d) Phantom read

95. For the given relation R(ABCDEF) and functional dependencies, find the correct candidate keys.
AB→C
DC→AE
E→F
a) {AB, BD, AD} b) {ABD, BCD} c) {BDE, BDF} d) {BD, AD, CD}

96. Which of the following best describes simulation in the context of discrete-event system modeling?
a) Exact mathematical solution of real-world problems
b) Replication of a system's behavior over time using a model
c) Optimization using deterministic techniques
d) Real-time experimentation on the actual system

97. Which statistical distribution is most often used to model interarrival times in a queuing system?
a) Normal distribution
b) Exponential distribution
c) Uniform distribution
d) Poisson distribution

98. The utilization factor (ρ) in a queuing system is defined as:
a) λ/μ b) μ/λ c) $1 - \lambda/\mu$ d) $\lambda \times \mu$

99. Which of the following techniques is used for selecting the "best-fit" distribution for simulation input data?
a) Kolmogorov–Smirnov test b) Bootstrapping
c) Monte Carlo test d) Random sampling

100. Verification in simulation refers to:
a) Comparing the simulation model output with real-world data
b) Ensuring the model has been correctly implemented according to specifications
c) Determining if the input distributions are realistic
d) Randomly testing system performance

VTU-ETR Seat No. | | | | | |

Question Paper Version : B

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) October - 2025

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PART – I

(Research Methodology)

[50 Marks]

1. The process of converting a normal random variable X to a standard normal variable Z is called:
a) Normalization | b) Standardization
c) Variance reduction | d) Centralization

2. In a Poisson distribution, the mean is equal to the:
a) Variance | b) Standard Deviation
c) Square of the variance | d) Probability of success

3. In mixed-method research design, which of the following best represents the "sequential explanatory strategy"?
a) Quantitative data collection followed by qualitative analysis for exploration
b) Qualitative exploration embedded within a quantitative experiment
c) Concurrent collection of qualitative and quantitative data with equal weight
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a) The problem is well defined
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5. In two-stage cluster sampling, the design effect (DEFF) is most directly influenced by:
a) Population mean | b) Intra-class correlation coefficient
c) Sampling frame completeness | d) Nonresponse rate

6. A statement such as, "If we increase our online advertising budget by 20%, then website traffic will increase by 15%" is an example of a:
a) Research objective | b) Management dilemma
c) Hypothesis | d) Research variable

7. The primary difference between a management dilemma and a research problem is that:
a) A management dilemma is faced by the researcher, while a research problem is faced by the manager.
b) A management dilemma is a symptom of an opportunity or problem, while a research problem is a statement about what information is needed to address it.
c) A management dilemma is always ethical in nature, while a research problem is methodological.
d) They are synonymous terms and can be used interchangeably.

8. A researcher is testing a new training program's effect on productivity. The possibility that the participants' awareness of being in an experiment itself influences their productivity is known as:
a) The halo effect | b) A contingency effect
c) An interactive testing effect | d) A surrogate information error

9. In the statement, "Brand loyalty is measured by the proportion of a customer's annual purchases in a category that are made from our brand," the phrase "proportion of purchases" serves as the:
a) Conceptual definition
b) Theoretical construct
c) Operational definition
d) Empirical scheme

10. A major fast-food chain wants to research the feasibility of launching a new plant-based burger nationwide. They decide to first conduct small-scale taste tests and focus groups in three cities. This initial study is an example of:
a) A conclusive research design
b) A cross-sectional study
c) Exploratory research
d) A performance-monitoring analysis

11. Which of the following illustrates a symptom rather than the true research problem?
a) Declining sales in a retail chain
b) Poor consumer perception of product quality
c) Ineffective promotional strategy
d) Limited market expansion opportunities

12. Which principle of the scientific method is most directly violated if a researcher hides unfavorable results?
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b) Replicable
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18. The theorem that revises prior probabilities based on new evidence is:
a) Central Limit Theorem
b) Bayes' Theorem
c) Theorem of Total Probability
d) Pythagoras Theorem

19. Which of the following is a necessary condition for a Binomial distribution?
a) The number of trials is variable.
b) Each trial has more than two possible outcomes.
c) The probability of success changes from trial to trial.
d) Trials are independent.

20. A situation where you count the number of heads in 20 coin tosses is best modeled by a:
a) Normal distribution b) Poisson distribution
c) Binomial distribution d) Uniform distribution

21. The main objective of a sample survey is to:
a) Collect data from every single member of the population
b) Collect data from a subset of the population to make inferences about the whole population
c) Only describe the characteristics of the sample itself
d) Replace the need for any statistical analysis

22. Which of the following is a probability sampling method?
a) Quota Sampling b) Judgmental Sampling
c) Simple Random Sampling d) Convenience Sampling

23. A sampling frame is:
a) The physical frame around a sample
b) The size of the sample
c) A list of all the elements in the population from which the sample is drawn
d) The margin of error in a survey

24. If a researcher stands in a shopping mall and selects shoppers who appear friendly to complete a survey, this is an example of:
a) Stratified Sampling b) Systematic Sampling
c) Snowball Sampling d) Convenience Sampling

25. The difference between the sample result and the true population value is known as:
a) Standard Deviation b) Sampling Error
c) Non-sampling Error d) Measurement Error

26. The process of checking completed questionnaires for accuracy, consistency, and completeness is called:
a) Data Analysis b) Data Editing
c) Data Coding d) Data Tabulation

27. Assigning numerical codes to response categories for data analysis is known as:
a) Data Editing b) Data Coding
c) Data Entry d) Data Validation

28. Which of the following is a method to handle missing data?
a) Ignore all cases with any missing values (listwise deletion)
b) Fabricate data to fill the gaps
c) Always replace missing values with the mean
d) There is only one correct method

29. The primary goal of data preparation is to:
a) Change the data to support the desired hypothesis
b) Ensure the data is accurate, consistent, and ready for analysis
c) Make the data look more impressive
d) Shorten the dataset by removing variables

30. Creating a frequency distribution table is part of which stage?
a) Data Collection
b) Data Presentation
c) Hypothesis Testing
d) Report Writing

31. What is the primary purpose of research?
a) To summarize existing knowledge
b) To discover new facts or verify old facts
c) To promote a specific political agenda
d) To create complex statistical models

32. Which of the following best describes 'Applied Research'?
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33. The first and most critical step in the research process is:
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a) Null hypothesis
b) Statistical hypothesis
c) Directional hypothesis
d) Complex hypothesis

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a) A printing error in a journal article
b) An area that has not been researched or is underexplored
c) The time gap between two published studies
d) The margin space on a printed page

41. A measure describing a characteristic of a population is called a:
a) Statistic b) Parameter
c) Sample d) Estimate

42. The Central Limit Theorem states that as sample size increases, the sampling distribution of the mean:
a) Becomes more skewed
b) Approaches a binomial distribution
c) Approaches a normal distribution, regardless of the population's distribution
d) Becomes identical to the population distribution

43. The standard deviation of the sampling distribution of a statistic is called its:
a) Standard Error b) Sampling Bias
c) Confidence Interval d) Population Deviation

44. An interval estimate, stated with a specific level of confidence, is called a:
a) Point Estimate b) Parameter Estimate
c) Confidence Interval d) Standard Interval

45. The purpose of statistical inference is to:
a) Describe the data in a sample
b) Make generalizations about a population from a sample
c) Collect data through surveys
d) Present data in graphical form

46. A hypothesis that is tested for possible rejection under the assumption that it is true is called the:
a) Alternative hypothesis b) Research hypothesis
c) Null hypothesis d) Directional hypothesis

47. The probability of rejecting a null hypothesis when it is actually true is known as:
a) Type II error (β) b) Level of Significance (α)
c) Power of the test ($1-\beta$) d) p-value

48. ANOVA is primarily used to test hypotheses about the differences among:
a) The variances of two populations
b) The means of two or more populations
c) The medians of two populations
d) The standard deviations of three populations

49. Which test is appropriate to compare means of two independent groups when population variances are equal and sample size moderately large?
a) Paired t-test　　b) Independent samples t-test (Student's t)
c) Chi-square test　　d) Wilcoxon signed-rank test

50 The "decision-making perspective" of business research emphasizes:
a) Maximizing the academic contribution of research
b) Reducing uncertainty for managers by providing relevant information
c) Collecting large amounts of data irrespective of its relevance
d) Promoting statistical software use in organizations

PART – II
Discipline Oriented Section
(Computer Science and Engineering/Information Science and Engineering)

[50 MARKS]

51. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____integrity constraint.
a) Referential b) Primary c) Referencing d) Specific

52. Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering
a) I only b) (B) II only c) Both I and II d) Neither I nor II

53. Which of the following concurrency control techniques does not ensure conflict-serializability?
a) Two-phase locking b) Timestamp ordering
c) Validation-based protocols d) Strict timestamp ordering with cascading aborts

54. Which of the following problems can still occur under the Read Committed isolation level?
a) Dirty read b) Lost update
c) Non-repeatable read d) Phantom read

55. For the given relation R(ABCDEF) and functional dependencies, find the correct candidate keys.
AB→C
DC→AE
E→F
a) {AB, BD, AD} b) {ABD, BCD} c) {BDE, BDF} d) {BD, AD, CD}

56. Which of the following best describes simulation in the context of discrete-event system modeling?
a) Exact mathematical solution of real-world problems
b) Replication of a system's behavior over time using a model
c) Optimization using deterministic techniques
d) Real-time experimentation on the actual system

57. Which statistical distribution is most often used to model interarrival times in a queuing system?
a) Normal distribution
b) Exponential distribution
c) Uniform distribution
d) Poisson distribution

58. The utilization factor (ρ) in a queuing system is defined as:
a) λ/μ b) μ/λ c) $1 - \lambda/\mu$ d) $\lambda \times \mu$

59. Which of the following techniques is used for selecting the "best-fit" distribution for simulation input data?
a) Kolmogorov–Smirnov test b) Bootstrapping
c) Monte Carlo test d) Random sampling

60. Verification in simulation refers to:
a) Comparing the simulation model output with real-world data
b) Ensuring the model has been correctly implemented according to specifications
c) Determining if the input distributions are realistic
d) Randomly testing system performance

61. In digital transmission, what is the major limitation of increasing baud rate beyond the channel bandwidth?
a) Signal attenuation increases linearly
b) Inter-symbol interference increases due to limited bandwidth
c) Noise decreases, improving signal quality
d) The data rate no longer applies

62. Which mechanism at the Transport Layer ensures reliability over an unreliable Network Layer?
a) Flow control and sequencing b) ARP and ICMP messages
c) Error detection at the physical layer d) CSMA/CD protocol

63. Which of the following is a characteristic of wired LANs such as Ethernet?
a) Uses CSMA/CD for collision detection b) Uses token passing for media access
c) Provides wireless connectivity d) Requires satellites for communication

64. Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer
c) Presentation layer d) Transport layer

65. Which of the following is true about the role of ARP (Address Resolution Protocol) in network communication?
a) ARP translates IP addresses to MAC addresses to facilitate communication on a LAN
b) ARP translates domain names to IP addresses
c) ARP is used to detect errors in transmitted packets
d) ARP provides routing information between different networks

66. A retail store system has classes Customer, Order, and Payment. If one Order must be linked to exactly one Payment, but a Customer can place many Orders, which multiplicities are correct?
a) Customer 1..* → Order 1..* ; Order 1..* → Payment 0..1
b) Customer 1 → Order 1..* ; Order 1 → Payment 1
c) Customer 1..* → Order 0..* ; Order 1 → Payment 0..*
d) Customer 1 → Order 0..* ; Order 1 → Payment 1

67. Which of the following best distinguishes use case diagrams from class diagrams in Object-Oriented Modelling?

a) Use case diagrams emphasize system structure while class diagrams emphasize system behavior

b) Use case diagrams emphasize functional requirements, while class diagrams emphasize static structure

c) Use case diagrams and class diagrams both model interactions with actors

d) Use case diagrams cannot describe relationships such as generalization

68. In a **ride-hailing system**, a *"Book Ride"* use case always requires *"Select Pickup Location"* and *"Select Drop Location"*, but only sometimes involves *"Apply Promo Code"*. Which UML modeling best represents this?

a) Extend for Pickup and Drop, Include for Promo Code
b) Include for Pickup and Drop, Extend for Promo Code
c) Generalization of all three
d) Association only

69. During use case realization, which diagram is typically used to **show the flow of messages between objects**?

a) Class Diagram b) Sequence Diagram
c) State Diagram d) Deployment Diagram

70. Which design pattern category does the *Observer pattern* belong to, and why?

a) Creational, because it dynamically generates observers on demand

b) Structural, because it defines object composition

c) Behavioral, because it manages interaction and communication among objects

d) Architectural, because it defines system-wide layering

71. Which of the following best defines software engineering?

a) The process of writing code to solve problems
b) Engineering discipline concerned with all aspects of software production
c) Management of software projects
d) None of the above

72. Which of the following is not one of the four major phases of RUP?

a) Inception b) Elaboration c) Construction d) Integration

73. In software project estimation, which of the following is not a valid size estimation technique?

a) Lines of Code (LOC) b) Function Point Analysis
c) Use-Case Points d) Data Flow Diagrams

74. What is the difference between functional and non-functional requirements?

a) Functional requirements define what the system should do, while non-functional requirements define how the system should behave

b) Functional requirements define how the system should behave, while non-functional requirements define what the system should do

c) Functional requirements are related to performance, while non-functional requirements are related to security

d) Functional requirements are related to security, while non-functional requirements are related to performance

75. In UML, aggregation represents:
a) Strong ownership (like inheritance)
b) A weak "whole-part" relationship where parts can exist independently
c) A dynamic interaction
d) Implementation of interfaces

76. Which of the following addressing modes allows the program to access a memory operand whose address is obtained by adding a constant value to the contents of a register?
a) Immediate addressing
b) Direct addressing
c) Indexed addressing
d) Register indirect addressing

77. Which of the following statements about cache memory is TRUE?
a) Cache memory is slower than main memory but cheaper
b) Cache memory uses magnetic storage technology
c) Cache memory eliminates the need for virtual memory
d) Cache memory stores a copy of a portion of main memory

78. A hardwired control unit is generally:
a) More flexible but slower than micro-programmed control
b) Faster but less flexible than micro-programmed control
c) Both slower and less flexible than micro-programmed control
d) Faster and more flexible than micro-programmed control

79. Direct Memory Access (DMA) improves performance because:
a) It reduces memory access time for the CPU
b) It allows I/O devices to access memory directly without CPU intervention
c) It eliminates the need for interrupts
d) It bypasses both CPU and memory in data transfer

80. In floating-point representation (IEEE single precision), the bias value for the exponent is:
a) 127
b) 63
c) 255
d) 1023

81. What is a major drawback of using recursion for problems that can also be solved iteratively?
a) Recursion always results in slower execution time
b) The risk of a stack overflow error for large inputs
c) The code is often more difficult to read and understand
d) It cannot handle problems with multiple base cases

82. In a linked list, what is the best approach to handle a memory leak caused by a failed deletion?
a) Not so important as modern systems handle memory management automatically
b) Use a ‘free()’ call on the node's pointer after it has been removed from the list's structure
c) Set the node's pointer to ‘NULL’ to indicate that it is no longer used
d) Introduce a garbage collection algorithm to track unreferenced nodes

83. A binary search tree is generated by inserting in the order of the following integers (from left to right). 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is
a) 4, 7
b) 8, 3
c) 7, 4
d) 6, 5

84 What will be the output of the following code?

```
int mystery(int n) {
   if (n <= 0) {
      return 0;
   } else {
      return n + mystery(n - 2);
   }
}

int main( ) {
printf("%d", mystery(5));
   return 0;
}
```

a) 0 b) 5 c) 9 d) 15

85. What is the maximum number of nodes in a binary tree of height 'h'?
a) 2h b) 2^h c) $2^{h+1}-1$ d) h^2

86. The truth values of the Propositions p, q and r if P →(Q V R) is False
a) P is True, Q is True and R is False
b) P is False, Q is False and R is True
c) P is True, Q is False and R is False
d) P is False, Q is True and R is True

87. How many integers between 1 and 300 (inclusive both) are divisible by at least one of 5, 6, 8?
a) 180 b) 120 c) 240 d) 138

88. In a directed Graph, there exists path between every pair of vertices, and then the Graph is called
a) Strongly connected Graph
b) Complete Graph
c) Isomorphic Graph
d) Euler Graph

89. A box contains 15 IC chips, of which 7 are defective and 8 are non- defective. In how many ways 5 chips can be chosen so that 2 are non-defective?
a) 56 b) 21 c) 1176 d) 980

90. Which one of the following Relation R: A-> A is not a function if A={1, 2, 3, 4}
a) R1={ (1, 2), (2, 4), (1, 3), (3, 4)}
b) R2={ (1, 1), (2, 2), (3, 3), (4, 4)}
c) R3={ (1, 3), (2, 4), (3, 4), (4, 1)}
d) R4={(1, 3), (2, 4, (3, 3), (4, 4)}

91. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is
a) $O(n^2)$ b) O (n long n) c) $O(n^2 \log n)$ d) $O(n^3)$

92. Which of the following is/are property/properties of a dynamic programming problem?
a) Optimal substructure
b) Overlapping sub problems
c) Greedy approach
d)Both optimal substructure and overlapping sub problems

93. What is the purpose of asymptotic notation in algorithm analysis?
a) To precisely measure the exact number of operations in an algorithm
b) To provide a rough estimate of algorithmic performance
c) To compare algorithms based on their exact execution times
d) To simplify the comparison of algorithms based on their growth rates

94. Which of the following algorithms is the best approach for solving Huffman codes?
a) Exhaustive Search
b) Greedy Algorithm
c) Brute Force Algorithm
d) Divide and Conquer Algorithm

95. Which of the following problems is NOT solved using dynamic programming?
a) 0/1 knapsack problem
b) Matrix chain multiplication problem
c) Edit distance problem
d) Fractional knapsack problem

96. Which of the following is an advantage of multi-threaded programming?
a) Increases context-switch overhead
b) Decreases responsiveness of applications
c) Improves responsiveness and resource sharing
d) Eliminates synchronization needs

97. A multi-threaded program has 5 user-level threads mapped to 2 kernel-level threads in a many-to-many model. If two threads issue blocking I/O calls, how many threads can still execute?
a) 5
b) 3
c) 2
d) 0

98. A process has a working set of 10 pages. The physical memory allocated is 8 frames. If the program accesses all 10 pages frequently:
a) No page faults will occur
b) Page faults may occur due to demand paging
c) Thrashing is likely to occur
d) The process will be terminated

99. Which of the following is true about user-level threads vs kernel-level threads?
a) User-level threads are slower to create and manage than kernel-level threads
b) Kernel-level threads are managed by the user-level library
c) User-level threads are scheduled by the user-level thread library
d) Kernel-level threads are invisible to the OS

100. Which of the following is stored in a page table?
a) Virtual address
b) Physical frame number
c) Page fault count
d) Process ID

VTU-ETR Seat No. | | | | | | |

Question Paper Version : C

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) October - 2025

Computer Science and Engineering/Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I

(Research Methodology)

[50 Marks]

1. Which of the following illustrates a symptom rather than the true research problem?
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c) The time gap between two published studies
d) The margin space on a printed page

21. The process of converting a normal random variable X to a standard normal variable Z is called:
a) Normalization
b) Standardization
c) Variance reduction
d) Centralization

22. In a Poisson distribution, the mean is equal to the:
a) Variance
b) Standard Deviation
c) Square of the variance
d) Probability of success

23. In mixed-method research design, which of the following best represents the "sequential explanatory strategy"?
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a) A management dilemma is faced by the researcher, while a research problem is faced by the manager.
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31. A measure describing a characteristic of a population is called a:
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c) Sample
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a) Becomes more skewed
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a) Describe the data in a sample
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a) The variances of two populations
b) The means of two or more populations
c) The medians of two populations
d) The standard deviations of three populations

39. Which test is appropriate to compare means of two independent groups when population variances are equal and sample size moderately large?
a) Paired t-test
b) Independent samples t-test (Student's t)
c) Chi-square test
d) Wilcoxon signed-rank test

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b) Reducing uncertainty for managers by providing relevant information
c) Collecting large amounts of data irrespective of its relevance
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c) Make the data look more impressive
d) Shorten the dataset by removing variables

50 Creating a frequency distribution table is part of which stage?
a) Data Collection b) Data Presentation
c) Hypothesis Testing d) Report Writing

PART – II
Discipline Oriented Section
(Computer Science and Engineering/Information Science and Engineering)

[50 MARKS]

51. In digital transmission, what is the major limitation of increasing baud rate beyond the channel bandwidth?
a) Signal attenuation increases linearly
b) Inter-symbol interference increases due to limited bandwidth
c) Noise decreases, improving signal quality
d) The data rate no longer applies

52. Which mechanism at the Transport Layer ensures reliability over an unreliable Network Layer?
a) Flow control and sequencing
b) ARP and ICMP messages
c) Error detection at the physical layer
d) CSMA/CD protocol

53. Which of the following is a characteristic of wired LANs such as Ethernet?
a) Uses CSMA/CD for collision detection
b) Uses token passing for media access
c) Provides wireless connectivity
d) Requires satellites for communication

54. Which layer in the OSI model is responsible for data presentation?
a) Network layer
b) Session layer
c) Presentation layer
d) Transport layer

55. Which of the following is true about the role of ARP (Address Resolution Protocol) in network communication?
a) ARP translates IP addresses to MAC addresses to facilitate communication on a LAN
b) ARP translates domain names to IP addresses
c) ARP is used to detect errors in transmitted packets
d) ARP provides routing information between different networks

56. A retail store system has classes Customer, Order, and Payment. If one Order must be linked to exactly one Payment, but a Customer can place many Orders, which multiplicities are correct?
a) Customer 1..* → Order 1..* ; Order 1..* → Payment 0..1
b) Customer 1 → Order 1..* ; Order 1 → Payment 1
c) Customer 1..* → Order 0..* ; Order 1 → Payment 0..*
d) Customer 1 → Order 0..* ; Order 1 → Payment 1

57. Which of the following best distinguishes use case diagrams from class diagrams in Object-Oriented Modelling?
a) Use case diagrams emphasize system structure while class diagrams emphasize system behavior
b) Use case diagrams emphasize functional requirements, while class diagrams emphasize static structure
c) Use case diagrams and class diagrams both model interactions with actors
d) Use case diagrams cannot describe relationships such as generalization

58. In a **ride-hailing system**, a *"Book Ride"* use case always requires *"Select Pickup Location"* and *"Select Drop Location"*, but only sometimes involves *"Apply Promo Code"*. Which UML modeling best represents this?
a) Extend for Pickup and Drop, Include for Promo Code
b) Include for Pickup and Drop, Extend for Promo Code
c) Generalization of all three
d) Association only

59. During use case realization, which diagram is typically used to **show the flow of messages between objects**?
a) Class Diagram
b) Sequence Diagram
c) State Diagram
d) Deployment Diagram

60. Which design pattern category does the *Observer pattern* belong to, and why?
a) Creational, because it dynamically generates observers on demand
b) Structural, because it defines object composition
c) Behavioral, because it manages interaction and communication among objects
d) Architectural, because it defines system-wide layering

61. What is a major drawback of using recursion for problems that can also be solved iteratively?
a) Recursion always results in slower execution time
b) The risk of a stack overflow error for large inputs
c) The code is often more difficult to read and understand
d) It cannot handle problems with multiple base cases

62. In a linked list, what is the best approach to handle a memory leak caused by a failed deletion?
a) Not so important as modern systems handle memory management automatically
b) Use a 'free()' call on the node's pointer after it has been removed from the list's structure
c) Set the node's pointer to 'NULL' to indicate that it is no longer used
d) Introduce a garbage collection algorithm to track unreferenced nodes

63. A binary search tree is generated by inserting in the order of the following integers (from left to right). 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is
a) 4, 7
b) 8, 3
c) 7, 4
d) 6, 5

64. What will be the output of the following code?

```
int mystery(int n) {
   if (n <= 0) {
      return 0;
   } else {
      return n + mystery(n - 2);
   }
}

int main( ) {
printf("%d", mystery(5));
   return 0;
}
```

a) 0
b) 5
c) 9
d) 15

65. What is the maximum number of nodes in a binary tree of height 'h'?
a) 2h b) 2^h c) $2^{h+1}-1$ d) h^2

66. The truth values of the Propositions p, q and r if P →(Q V R) is False
a) P is True, Q is True and R is False
b) P is False, Q is False and R is True
c) P is True, Q is False and R is False
d) P is False, Q is True and R is True

67. How many integers between 1 and 300 (inclusive both) are divisible by at least one of 5, 6, 8?
a) 180 b) 120 c) 240 d) 138

68. In a directed Graph, there exists path between every pair of vertices, and then the Graph is called
a) Strongly connected Graph
b) Complete Graph
c) Isomorphic Graph
d) Euler Graph

69. A box contains 15 IC chips, of which 7 are defective and 8 are non- defective. In how many ways 5 chips can be chosen so that 2 are non-defective?
a) 56 b) 21 c) 1176 d) 980

70. Which one of the following Relation R: A-> A is not a function if A={1, 2, 3, 4}
a) R1={ (1, 2), (2, 4), (1, 3), (3, 4)}
b) R2={ (1, 1), (2, 2), (3, 3), (4, 4)}
c) R3={ (1, 3), (2, 4), (3, 4), (4, 1)}
d) R4={(1, 3), (2, 4, (3, 3), (4, 4)}

71. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____integrity constraint.
a) Referential b) Primary c) Referencing d) Specific

72. Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering
a) I only b) (B) II only c) Both I and II d) Neither I nor II

73. Which of the following concurrency control techniques does not ensure conflict-serializability?
a) Two-phase locking
b) Timestamp ordering
c) Validation-based protocols
d) Strict timestamp ordering with cascading aborts

74. Which of the following problems can still occur under the Read Committed isolation level?
a) Dirty read
b) Lost update
c) Non-repeatable read
d) Phantom read

75. For the given relation R(ABCDEF) and functional dependencies, find the correct candidate keys.
AB→C
DC→AE
E→F
a) {AB, BD, AD} b) {ABD, BCD} c) {BDE, BDF} d) {BD, AD, CD}

76. Which of the following best describes simulation in the context of discrete-event system modeling?
a) Exact mathematical solution of real-world problems
b) Replication of a system's behavior over time using a model
c) Optimization using deterministic techniques
d) Real-time experimentation on the actual system

77. Which statistical distribution is most often used to model interarrival times in a queuing system?
a) Normal distribution
b) Exponential distribution
c) Uniform distribution
d) Poisson distribution

78. The utilization factor (ρ) in a queuing system is defined as:
a) λ/μ b) μ/λ c) $1 - \lambda/\mu$ d) $\lambda \times \mu$

79. Which of the following techniques is used for selecting the "best-fit" distribution for simulation input data?
a) Kolmogorov–Smirnov test b) Bootstrapping
c) Monte Carlo test d) Random sampling

80. Verification in simulation refers to:
a) Comparing the simulation model output with real-world data
b) Ensuring the model has been correctly implemented according to specifications
c) Determining if the input distributions are realistic
d) Randomly testing system performance

81. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is
a) $O(n^2)$ b) O (n long n) c) $O(n^2 \log n)$ d) $O(n^3)$

82. Which of the following is/are property/properties of a dynamic programming problem?
a) Optimal substructure
b) Overlapping sub problems
c) Greedy approach
d)Both optimal substructure and overlapping sub problems

83. What is the purpose of asymptotic notation in algorithm analysis?
a) To precisely measure the exact number of operations in an algorithm
b) To provide a rough estimate of algorithmic performance
c) To compare algorithms based on their exact execution times
d) To simplify the comparison of algorithms based on their growth rates

84 Which of the following algorithms is the best approach for solving Huffman codes?
a) Exhaustive Search
b) Greedy Algorithm
c) Brute Force Algorithm
d) Divide and Conquer Algorithm

85. Which of the following problems is NOT solved using dynamic programming?
a) 0/1 knapsack problem
b) Matrix chain multiplication problem
c) Edit distance problem
d) Fractional knapsack problem

86. Which of the following is an advantage of multi-threaded programming?
a) Increases context-switch overhead
b) Decreases responsiveness of applications
c) Improves responsiveness and resource sharing
d) Eliminates synchronization needs

87. A multi-threaded program has 5 user-level threads mapped to 2 kernel-level threads in a many-to-many model. If two threads issue blocking I/O calls, how many threads can still execute?
a) 5
b) 3
c) 2
d) 0

88. A process has a working set of 10 pages. The physical memory allocated is 8 frames. If the program accesses all 10 pages frequently:
a) No page faults will occur
b) Page faults may occur due to demand paging
c) Thrashing is likely to occur
d) The process will be terminated

89. Which of the following is true about user-level threads vs kernel-level threads?
a) User-level threads are slower to create and manage than kernel-level threads
b) Kernel-level threads are managed by the user-level library
c) User-level threads are scheduled by the user-level thread library
d) Kernel-level threads are invisible to the OS

90. Which of the following is stored in a page table?
a) Virtual address
b) Physical frame number
c) Page fault count
d) Process ID

91. Which of the following best defines software engineering?
a) The process of writing code to solve problems
b) Engineering discipline concerned with all aspects of software production
c) Management of software projects
d) None of the above

92. Which of the following is not one of the four major phases of RUP?
a) Inception
b) Elaboration
c) Construction
d) Integration

93. In software project estimation, which of the following is not a valid size estimation technique?
a) Lines of Code (LOC)
b) Function Point Analysis
c) Use-Case Points
d) Data Flow Diagrams

94. What is the difference between functional and non-functional requirements?

a) Functional requirements define what the system should do, while non-functional requirements define how the system should behave

b) Functional requirements define how the system should behave, while non-functional requirements define what the system should do

c) Functional requirements are related to performance, while non-functional requirements are related to security

d) Functional requirements are related to security, while non-functional requirements are related to performance

95. In UML, aggregation represents:
a) Strong ownership (like inheritance)
b) A weak "whole-part" relationship where parts can exist independently
c) A dynamic interaction
d) Implementation of interfaces

96. Which of the following addressing modes allows the program to access a memory operand whose address is obtained by adding a constant value to the contents of a register?
a) Immediate addressing
b) Direct addressing
c) Indexed addressing
d) Register indirect addressing

97. Which of the following statements about cache memory is TRUE?
a) Cache memory is slower than main memory but cheaper
b) Cache memory uses magnetic storage technology
c) Cache memory eliminates the need for virtual memory
d) Cache memory stores a copy of a portion of main memory

98. A hardwired control unit is generally:
a) More flexible but slower than micro-programmed control
b) Faster but less flexible than micro-programmed control
c) Both slower and less flexible than micro-programmed control
d) Faster and more flexible than micro-programmed control

99. Direct Memory Access (DMA) improves performance because:
a) It reduces memory access time for the CPU
b) It allows I/O devices to access memory directly without CPU intervention
c) It eliminates the need for interrupts
d) It bypasses both CPU and memory in data transfer

100. In floating-point representation (IEEE single precision), the bias value for the exponent is:
a) 127
b) 63
c) 255
d) 1023

VTU-ETR Seat No. | | | | | | |

Question Paper Version : D

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research) October - 2025

DO NOT USE MOBILE PHONES

Computer Science and Engineering/Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

PART – I

(Research Methodology)

[50 Marks]

1. A measure describing a characteristic of a population is called a:
a) Statistic
b) Parameter
c) Sample
d) Estimate

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11. The process of converting a normal random variable X to a standard normal variable Z is called:
a) Normalization
b) Standardization
c) Variance reduction
d) Centralization

12. In a Poisson distribution, the mean is equal to the:
a) Variance
b) Standard Deviation
c) Square of the variance
d) Probability of success

13. In mixed-method research design, which of the following best represents the "sequential explanatory strategy"?
a) Quantitative data collection followed by qualitative analysis for exploration
b) Qualitative exploration embedded within a quantitative experiment
c) Concurrent collection of qualitative and quantitative data with equal weight
d) Iterative cycles of qualitative and quantitative methods without fixed precedence

14. Exploratory research is usually conducted when:
a) The problem is well defined
b) The researcher has little knowledge about the problem
c) The study is about testing a clear hypothesis
d) The goal is to make accurate forecasts

15. In two-stage cluster sampling, the design effect (DEFF) is most directly influenced by:
a) Population mean
b) Intra-class correlation coefficient
c) Sampling frame completeness
d) Nonresponse rate

16. A statement such as, "If we increase our online advertising budget by 20%, then website traffic will increase by 15%" is an example of a:
a) Research objective
b) Management dilemma
c) Hypothesis
d) Research variable

17. The primary difference between a management dilemma and a research problem is that:
a) A management dilemma is faced by the researcher, while a research problem is faced by the manager.
b) A management dilemma is a symptom of an opportunity or problem, while a research problem is a statement about what information is needed to address it.
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21. Which of the following illustrates a symptom rather than the true research problem?
a) Declining sales in a retail chain
b) Poor consumer perception of product quality
c) Ineffective promotional strategy
d) Limited market expansion opportunities

22. Which principle of the scientific method is most directly violated if a researcher hides unfavorable results?
a) Replication
b) Objectivity
c) Empiricism
d) Parsimony

23. Which of the following is considered an ethical responsibility of a business researcher?
a) Guaranteeing positive results for the client
b) Maintaining respondent confidentiality
c) Concealing the research methodology to protect proprietary techniques
d) Ensuring sponsors dictate all research questions

24. Which of the following is an example of secondary data?
a) A company's sales reports from last year
b) A new survey of customer satisfaction
c) A focus group discussion with consumers
d) An experiment testing price sensitivity

25. If an ANOVA test yields a significant result, it indicates that:
a) All group means are significantly different from each other.
b) At least one group mean is significantly different from the others.
c) The variances of the groups are equal.
d) The null hypothesis of equal means is accepted.

26. In simple random sampling (SRS) without replacement, increasing sample size will generally:
a) Increase standard error of an estimate
b) Decrease standard error of an estimate
c) Not affect standard error
d) Make estimates invalid

27. Which of these is NOT a characteristic of good research?
a) Systematic
b) Replicable
c) Biased conclusions to support the investigator
d) Logical and empirical

28. The theorem that revises prior probabilities based on new evidence is:
a) Central Limit Theorem
b) Bayes' Theorem
c) Theorem of Total Probability
d) Pythagoras Theorem

29. Which of the following is a necessary condition for a Binomial distribution?
a) The number of trials is variable.
b) Each trial has more than two possible outcomes.
c) The probability of success changes from trial to trial.
d) Trials are independent.

30. A situation where you count the number of heads in 20 coin tosses is best modeled by a:
a) Normal distribution b) Poisson distribution
c) Binomial distribution d) Uniform distribution

31. The main objective of a sample survey is to:
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d) Report Writing

41. What is the primary purpose of research?
a) To summarize existing knowledge
b) To discover new facts or verify old facts
c) To promote a specific political agenda
d) To create complex statistical models

42. Which of the following best describes 'Applied Research'?
a) Research carried out to generate knowledge for its own sake.
b) Research aimed at finding a solution to an immediate problem
c) Research that only uses qualitative methods
d) Research based purely on historical data.

43. The first and most critical step in the research process is:
a) Formulating a hypothesis
b) Defining the research problem
c) Collecting data
d) Preparing the research report

44. A hypothesis that indicates a direction of the expected difference or relationship is known as a:
a) Null hypothesis
b) Statistical hypothesis
c) Directional hypothesis
d) Complex hypothesis

45. Which research design is primarily used to establish cause-and-effect relationships?
a) Descriptive research design
b) Exploratory research design
c) Diagnostic research design
d) Experimental research design

46. What is the primary purpose of a literature review in research?
a) To show off how many books you have read
b) To avoid having to do original research
c) To gain insights into the existing theoretical and research background of the problem
d) To copy the methodology of previous studies exactly

47. A literature review should primarily be based on:
a) Personal opinions and anecdotes
b) Secondary sources like textbooks and review articles
c) Primary sources like original research reports and journal articles
d) Unverified online blogs and websites

48. Which of the following is a key characteristic of a good literature review?
a) It is highly subjective and opinionated.
b) It is a simple list of summaries of various articles.
c) It is comprehensive, critical, and synthesizes existing work.
d) It only includes studies that support the researcher's preconceived notions.

49. A 'secondary source' in a literature review refers to:
a) The second most important source you find
b) An interpretation or analysis of primary sources by other authors
c) The original reporting of research by the person who conducted it
d) A source that is less than 5 years old

50 The 'gap in literature' that a researcher identifies refers to:
a) A printing error in a journal article
b) An area that has not been researched or is underexplored
c) The time gap between two published studies
d) The margin space on a printed page

PART – II
Discipline Oriented Section
(Computer Science and Engineering/Information Science and Engineering)

[50 MARKS]

51. You have an array of n elements. Suppose you implement quick sort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is
a) $O(n^2)$ b) O (n long n) c) $O(n^2 \log n)$ d) $O(n^3)$

52. Which of the following is/are property/properties of a dynamic programming problem?
a) Optimal substructure
b) Overlapping sub problems
c) Greedy approach
d)Both optimal substructure and overlapping sub problems

53. What is the purpose of asymptotic notation in algorithm analysis?
a) To precisely measure the exact number of operations in an algorithm
b) To provide a rough estimate of algorithmic performance
c) To compare algorithms based on their exact execution times
d) To simplify the comparison of algorithms based on their growth rates

54. Which of the following algorithms is the best approach for solving Huffman codes?
a) Exhaustive Search
b) Greedy Algorithm
c) Brute Force Algorithm
d) Divide and Conquer Algorithm

55. Which of the following problems is NOT solved using dynamic programming?
a) 0/1 knapsack problem b) Matrix chain multiplication problem
c) Edit distance problem d) Fractional knapsack problem

56. Which of the following is an advantage of multi-threaded programming?
a) Increases context-switch overhead
b) Decreases responsiveness of applications
c) Improves responsiveness and resource sharing
d) Eliminates synchronization needs

57. A multi-threaded program has 5 user-level threads mapped to 2 kernel-level threads in a many-to-many model. If two threads issue blocking I/O calls, how many threads can still execute?
a) 5 b) 3 c) 2 d) 0

58. A process has a working set of 10 pages. The physical memory allocated is 8 frames. If the program accesses all 10 pages frequently:
a) No page faults will occur
b) Page faults may occur due to demand paging
c) Thrashing is likely to occur
d) The process will be terminated

59. Which of the following is true about user-level threads vs kernel-level threads?
a) User-level threads are slower to create and manage than kernel-level threads
b) Kernel-level threads are managed by the user-level library
c) User-level threads are scheduled by the user-level thread library
d) Kernel-level threads are invisible to the OS

60. Which of the following is stored in a page table?
a) Virtual address b) Physical frame number
c) Page fault count d) Process ID

61. The values appearing in given attributes of any tuple in the referencing relation must likewise occur in specified attributes of at least one tuple in the referenced relation, according to _____integrity constraint.
a) Referential b) Primary c) Referencing d) Specific

62. Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?
I. 2-phase locking
II. Time-stamp ordering
a) I only b) (B) II only c) Both I and II d) Neither I nor II

63. Which of the following concurrency control techniques does not ensure conflict-serializability?
a) Two-phase locking b) Timestamp ordering
c) Validation-based protocols d) Strict timestamp ordering with cascading aborts

64. Which of the following problems can still occur under the Read Committed isolation level?
a) Dirty read b) Lost update
c) Non-repeatable read d) Phantom read

65. For the given relation R(ABCDEF) and functional dependencies, find the correct candidate keys.
$AB \rightarrow C$
$DC \rightarrow AE$
$E \rightarrow F$
a) {AB, BD, AD} b) {ABD, BCD} c) {BDE, BDF} d) {BD, AD, CD}

66. Which of the following best describes simulation in the context of discrete-event system modeling?
a) Exact mathematical solution of real-world problems
b) Replication of a system's behavior over time using a model
c) Optimization using deterministic techniques
d) Real-time experimentation on the actual system

67. Which statistical distribution is most often used to model interarrival times in a queuing system?
a) Normal distribution
b) Exponential distribution
c) Uniform distribution
d) Poisson distribution

68. The utilization factor (ρ) in a queuing system is defined as:
a) λ/μ b) μ/λ c) $1 - \lambda/\mu$ d) $\lambda \times \mu$

69. Which of the following techniques is used for selecting the "best-fit" distribution for simulation input data?
a) Kolmogorov–Smirnov test b) Bootstrapping
c) Monte Carlo test d) Random sampling

70. Verification in simulation refers to:
a) Comparing the simulation model output with real-world data
b) Ensuring the model has been correctly implemented according to specifications
c) Determining if the input distributions are realistic
d) Randomly testing system performance

71. In digital transmission, what is the major limitation of increasing baud rate beyond the channel bandwidth?
a) Signal attenuation increases linearly
b) Inter-symbol interference increases due to limited bandwidth
c) Noise decreases, improving signal quality
d) The data rate no longer applies

72. Which mechanism at the Transport Layer ensures reliability over an unreliable Network Layer?
a) Flow control and sequencing b) ARP and ICMP messages
c) Error detection at the physical layer d) CSMA/CD protocol

73. Which of the following is a characteristic of wired LANs such as Ethernet?
a) Uses CSMA/CD for collision detection b) Uses token passing for media access
c) Provides wireless connectivity d) Requires satellites for communication

74. Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer
c) Presentation layer d) Transport layer

75. Which of the following is true about the role of ARP (Address Resolution Protocol) in network communication?
a) ARP translates IP addresses to MAC addresses to facilitate communication on a LAN
b) ARP translates domain names to IP addresses
c) ARP is used to detect errors in transmitted packets
d) ARP provides routing information between different networks

76. A retail store system has classes Customer, Order, and Payment. If one Order must be linked to exactly one Payment, but a Customer can place many Orders, which multiplicities are correct?
a) Customer 1..* → Order 1..* ; Order 1..* → Payment 0..1
b) Customer 1 → Order 1..* ; Order 1 → Payment 1
c) Customer 1..* → Order 0..* ; Order 1 → Payment 0..*
d) Customer 1 → Order 0..* ; Order 1 → Payment 1

77. Which of the following best distinguishes use case diagrams from class diagrams in Object-Oriented Modelling?

a) Use case diagrams emphasize system structure while class diagrams emphasize system behavior

b) Use case diagrams emphasize functional requirements, while class diagrams emphasize static structure

c) Use case diagrams and class diagrams both model interactions with actors

d) Use case diagrams cannot describe relationships such as generalization

78. In a **ride-hailing system**, a *"Book Ride"* use case always requires *"Select Pickup Location"* and *"Select Drop Location"*, but only sometimes involves *"Apply Promo Code"*. Which UML modeling best represents this?

a) Extend for Pickup and Drop, Include for Promo Code
b) Include for Pickup and Drop, Extend for Promo Code
c) Generalization of all three
d) Association only

79. During use case realization, which diagram is typically used to **show the flow of messages between objects**?

a) Class Diagram
b) Sequence Diagram
c) State Diagram
d) Deployment Diagram

80. Which design pattern category does the *Observer pattern* belong to, and why?

a) Creational, because it dynamically generates observers on demand

b) Structural, because it defines object composition

c) Behavioral, because it manages interaction and communication among objects

d) Architectural, because it defines system-wide layering

81. Which of the following best defines software engineering?

a) The process of writing code to solve problems
b) Engineering discipline concerned with all aspects of software production
c) Management of software projects
d) None of the above

82. Which of the following is not one of the four major phases of RUP?

a) Inception
b) Elaboration
c) Construction
d) Integration

83. In software project estimation, which of the following is not a valid size estimation technique?

a) Lines of Code (LOC)
b) Function Point Analysis
c) Use-Case Points
d) Data Flow Diagrams

84 What is the difference between functional and non-functional requirements?

a) Functional requirements define what the system should do, while non-functional requirements define how the system should behave

b) Functional requirements define how the system should behave, while non-functional requirements define what the system should do

c) Functional requirements are related to performance, while non-functional requirements are related to security

d) Functional requirements are related to security, while non-functional requirements are related to performance

85. In UML, aggregation represents:
a) Strong ownership (like inheritance)
b) A weak "whole-part" relationship where parts can exist independently
c) A dynamic interaction
d) Implementation of interfaces

86. Which of the following addressing modes allows the program to access a memory operand whose address is obtained by adding a constant value to the contents of a register?
a) Immediate addressing
b) Direct addressing
c) Indexed addressing
d) Register indirect addressing

87. Which of the following statements about cache memory is TRUE?
a) Cache memory is slower than main memory but cheaper
b) Cache memory uses magnetic storage technology
c) Cache memory eliminates the need for virtual memory
d) Cache memory stores a copy of a portion of main memory

88. A hardwired control unit is generally:
a) More flexible but slower than micro-programmed control
b) Faster but less flexible than micro-programmed control
c) Both slower and less flexible than micro-programmed control
d) Faster and more flexible than micro-programmed control

89. Direct Memory Access (DMA) improves performance because:
a) It reduces memory access time for the CPU
b) It allows I/O devices to access memory directly without CPU intervention
c) It eliminates the need for interrupts
d) It bypasses both CPU and memory in data transfer

90. In floating-point representation (IEEE single precision), the bias value for the exponent is:
a) 127
b) 63
c) 255
d) 1023

91. What is a major drawback of using recursion for problems that can also be solved iteratively?
a) Recursion always results in slower execution time
b) The risk of a stack overflow error for large inputs
c) The code is often more difficult to read and understand
d) It cannot handle problems with multiple base cases

92. In a linked list, what is the best approach to handle a memory leak caused by a failed deletion?
a) Not so important as modern systems handle memory management automatically
b) Use a ‘free()’ call on the node's pointer after it has been removed from the list's structure
c) Set the node's pointer to ‘NULL’ to indicate that it is no longer used
d) Introduce a garbage collection algorithm to track unreferenced nodes

93. A binary search tree is generated by inserting in the order of the following integers (from left to right). 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. The number of nodes in the left subtree and right subtree of the root respectively is
a) 4, 7
b) 8, 3
c) 7, 4
d) 6, 5

94. What will be the output of the following code?

```
int mystery(int n) {
   if (n <= 0) {
      return 0;
   } else {
      return n + mystery(n - 2);
   }
}

int main( ) {
printf("%d", mystery(5));
   return 0;
}
```

a) 0 b) 5 c) 9 d) 15

95. What is the maximum number of nodes in a binary tree of height 'h'?

a) 2h b) 2^h c) $2^{h+1}-1$ d) h^2

96. The truth values of the Propositions p, q and r if P →(Q V R) is False

a) P is True, Q is True and R is False b) P is False, Q is False and R is True
c) P is True, Q is False and R is False d) P is False, Q is True and R is True

97. How many integers between 1 and 300 (inclusive both) are divisible by at least one of 5, 6, 8?

a) 180 b) 120 c) 240 d) 138

98. In a directed Graph, there exists path between every pair of vertices, and then the Graph is called

a) Strongly connected Graph b) Complete Graph
c) Isomorphic Graph d) Euler Graph

99. A box contains 15 IC chips, of which 7 are defective and 8 are non- defective. In how many ways 5 chips can be chosen so that 2 are non-defective?

a) 56 b) 21 c) 1176 d) 980

100. Which one of the following Relation R: A-> A is not a function if A={1, 2, 3, 4}

a) R1={ (1, 2), (2, 4), (1, 3), (3, 4)} b) R2={ (1, 1), (2, 2), (3, 3), (4, 4)}
c) R3={ (1, 3), (2, 4), (3, 4), (4, 1)} d) R4={(1, 3), (2, 4, (3, 3), (4, 4)}